THIRD SEMESTER MGU-UGP (HONOURS) DEGREE REGULAR EXAMINATION, OCTOBER 2025

MG3DSCCFS2001- COMPUTER ORGANIZATION

SCHEME

QP CODE: 25900333 **Max Marks-50**

PART A (Answer any 10)

1. Memory Address Register (MAR) holds the address of the memory location to be accessed.

2. Conditional branch is executed only if a specific condition is true.

3. Control Unit directs the operation of the processor.

4. Bus is a common communication path for transferring data among system components.

5. Program Counter (PC) holds the address of the next instruction to be executed.

6. Stack Pointer (SP) points to the top of the stack in memory.

7. Auxiliary memory refers to secondary storage devices like hard disks.

8. Hard disks are preferred for OS and applications because they are non-volatile and offer large capacity.

9. Two types of cache memory are Level 1 (L1) and Level 2 (L2) cache.

10. ROM stores the bootstrap program used for system booting.

11. Cache memory reduces access time between CPU and main memory, improving performance.

12. DRAM is used for large memory modules due to its high density and low cost per bit.

PART B (Answer any 4)

13. Load operation transfers data from memory to register, while Store transfers data from register to memory.

14. CPU Registers include PC, IR, MAR, MDR, AC, SP, and general-purpose registers. Each performs control, storage, or execution-related tasks.

15. Auto-increment addressing mode automatically increments the register after accessing data, useful in array and string operations.

16. Relative addressing add the offset to PC; Indexed addressing add the content of index register to the address part of instruction. Relative is used for branching, Indexed for arrays.

17. Processing Unit executes instructions, performs arithmetic/logical operations, transfers data, and coordinates CPU activities.

18. Instruction Register (IR) holds the instruction being executed, helping the control unit decode and execute correctly.

PART C (Answer any 2)

19. Functional units of a computer include Input Unit, Output Unit, Memory Unit, ALU, and Control Unit. Input unit accepts data; memory stores data and programs; ALU performs arithmetic and logic operations; Control Unit manages all operations; Output unit displays results. Data flow: Input → Memory → CPU → Output.

20. Data transfer and manipulation instructions include MOV, LOAD, STORE, PUSH, POP for transfer; ADD, SUB, AND, OR, SHIFT, ROTATE for manipulation. These operations move, modify, and process data efficiently within CPU cycles (fetch, decode, execute).

21. Arithmetic operations like ADD, SUB, MUL, DIV perform calculations, while logic operations like AND, OR, XOR, NOT work on bits. Arithmetic affects numerical results; logic changes bit patterns. Example: ADD R1, R2 vs AND R1, R2.